

Personal Protective Equipment



Insulated
Gloves



Lab Coat



Scooper



Safety
Goggles*

* If splash
potential
exists, wear
goggles + face
shield

What is Dry Ice?

Dry ice is carbon dioxide in solid form. As it warms, dry ice turns to a gas state (sublimates), releasing carbon dioxide. Carbon dioxide is heavier than air and can displace oxygen and accumulate in closed spaces, creating an oxygen-deficient environment.

Hazards & Best Practices



Frostbite/burns: Dry ice can cause severe frostbite within seconds of skin contact. Never handle dry ice with bare hands. Always use personal protective equipment.



Physical: Always open the dry ice freezer lid to maximum opening. Lids that are not secured can cause injury.



Explosion: Store dry ice in a container that allows release of gas such as a vented or Styrofoam cooler. Placing dry ice inside an enclosed space, such as an ultra-low temperature freezer, or a tightly sealed container could result in rupture.



Suffocation: Avoid inhalation. Always store and handle dry ice in a well-ventilated room. Maintain distance between yourself and the dry ice by using a scooper with a long handle. Do not lean into the dry ice tote or hover over oxygen deficient area.

Reminders

- Do not dispose of dry ice in a sink drain or toilet as pipes can freeze and rupture. Place the dry ice in an open container in a chemical fume hood or well-ventilated room to sublimate.
- Required training for personnel working with dry ice include:
 - Cryogenic Liquids and Dry Ice Safety (EHS-2480)
 - DOT Shipping Dangerous Biological Goods or Dry Ice (EHS-2700) for those shipping dry ice; Refresher (EHS-2701) every two years.
- Always store metal scoopers outside of the freezer.
- Be familiar with incident response and reporting steps (stanford.io/3pvwof8).

First Aid

- If the individual is experiencing any difficulty breathing, remove them to fresh air. Seek medical attention immediately if needed. For skin or mucous membrane exposure, immediately rinse eyes/skin thoroughly with water for 15 minutes in the eyewash/safety shower.
- Consult the Occupational Health Center at (650) 725-5308 for guidance on first aid.
- Refer to the Safety Data Sheet (SDS) for additional information.